

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

CHEMISTRY

Paper 1 Multiple Choice

8872/01

FRIDAY 16 SEPTEMBER 2011

50 Minutes

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write and/or shade your name, NRIC / FIN number and HT group on the Multiple Choice Answer Sheet provided.

There are **30 MCQ** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 In an attempt to establish the formula of an oxide of nitrogen, a known volume of the pure gas, N_xO_y was converted to ammonia and water according to the following equation:



2400 cm³ of the oxide of nitrogen, measured at room temperature and pressure (r.t.p.) produced 7.20 g of water. The ammonia produced was neutralised by 200 cm³ of 1.0 mol dm⁻³ HCl. What is the formula of the oxide of nitrogen?

- A** NO **B** NO₂ **C** N₂O **D** N₂O₄

$$\text{No. of moles of oxide of nitrogen} = \frac{2400}{24000} = 0.100$$

$$\text{No. of moles of water} = \frac{7.20}{18.0} = 0.400$$

$$\text{No. of moles of HCl} = \text{no. of moles of ammonia} = \frac{200}{1000} \times 1.0 = 0.200$$

So, $N_xO_y : NH_3 : H_2O$ is 1:2:4. Hence, $x = 2$ and $y = 4$.
Oxide of nitrogen is N₂O₄.

- 2 The relative atomic mass of boron, which consists of the isotopes ¹⁰B and ¹¹B is 10.8. What is the percentage ¹¹B atoms in the isotopic mixture?

- A** 0.8% **B** 8.0% **C** 20% **D** 80%

Let x be the fraction of ¹¹B in the mixture.

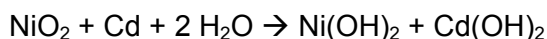
So $(1-x)$ is the fraction of ¹⁰B in the mixture.

$$11x + (1-x)10 = 10.8$$

$$x = 0.8.$$

$$\text{Percentage } ^{11}\text{B atoms in the isotopic mixture} = 0.8 \times 100\% = 80\%.$$

- 3 Nickel-Cadmium rechargeable batteries were once commonly used in handphones and electric shavers. The reaction that occurs when the battery is used is:



When the battery goes flat, it can be recharged and the reaction is reversed, restoring the NiO₂ and Cd.

What is the change in the oxidation state of nickel when the battery is being used?

- A** from +4 to +2
B from +2 to +4
C from +2 to +1
D from +4 to +1

In NiO₂, oxygen has an oxidation state of -2. So, overall oxidation charge for oxygen is $2 \times (-2)$ which is -4. Hence, Ni has an oxidation state of +4 since NiO₂ is neutral. In Ni(OH)₂, OH⁻ has charge of -1 and two of that will have a charge of -2. So Ni will have to be +2.

- 4 Which of the following electronic configurations represents an element that most likely forms a simple ion with a charge of -3?
- A $1s^2 2s^2 2p^6 3s^2 3p^1$
- B $1s^2 2s^2 2p^6 3s^2 3p^3$**
- C $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$
- D $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$

A charge of -3 indicates that the element is likely to be a non-metal. That rules out options C and D which are the transition metals (metals are elements that have partially filled *d* orbitals).

Option A is aluminium which is more likely to form a simple ion with a charge of +3.

B is the correct answer (phosphorus) as it will likely to gain 3 electrons to completely fill the 3p subshell and hence attain a charge of -3.

- 5 Use of the Data Booklet is relevant to this question.
The successive ionisation energies, in kJ mol^{-1} , of an element X is given below:

870 1800 3000 3600 5800 7000 13200

What is the identity of X?

- A ${}_8\text{O}$ B ${}_{33}\text{As}$ **C ${}_{52}\text{Te}$** D ${}_{53}\text{I}$

The sharp increase in ionisation energy is between 7000 to 13200 kJ mol^{-1} , that is between the 6th and 7th IE. Hence, X has to be in group VI. Answer is either A or C. Upon referring to the ionisation energy values in data booklet, X cannot be oxygen, so the answer is Te (which has lower IE than O due to shielding effect and distance of valence electron from the nucleus.)

- 6 In which one of the following pairs do the molecules have similar shapes?

- A AlCl_3 and PCl_3
- B AlCl_3 and BF_3**
- C BeCl_2 and H_2O
- D BF_3 and NH_3

A AlCl_3 , BF_3 : trigonal planar

B PCl_3 , NH_3 : trigonal pyramidal due to lone pair on P and N respectively.

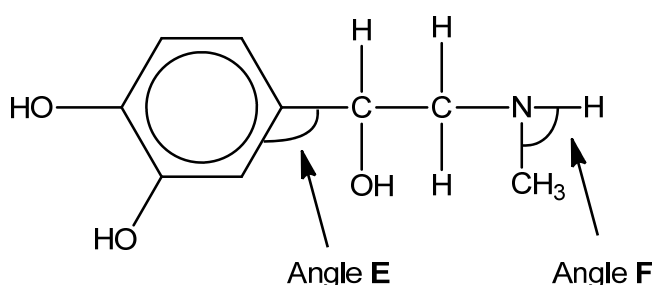
C BeCl_2 is linear.

D H_2O is bent due to the two lone pairs of electrons on O.

- 7 In which one of the following pairs of species is the melting point of the first member lower than that of the second member?
- A diamond, graphite
- B H_2O , NH_3
- C Br_2 , ICl**
- D pentane, 2,2-dimethylpropane

- A** The melting of diamond requires the breaking of 4 covalent bonds per carbon atoms as opposed to graphite where only 3 covalent bonds per carbon atom. Weak van der waals' forces are broken in between the layers of graphite.
- B** H₂O has more extensive hydrogen bonding than NH₃. This is evident by the fact that H₂O is liquid at room temperature and pressure while NH₃ is a gas.
- C** Br₂ is non-polar and hence the intermolecular forces broken are van der waal's forces. These are weaker than the intermolecular dipole-dipole forces between the polar ICl molecules.
- D** Pentane and 2,2-dimethylpropane are structural isomers(chain isomerism). Both pentane and 2,2-dimethylpropane are non-polar molecules that forms van de waal's forces, but 2,2-dimethylpropane has a smaller surface area for van de waal's forces to form between molecules.

- 8** Adrenaline is a hormone which, when secreted directly into the blood stream, acts as a stimulant. It has the following structure:

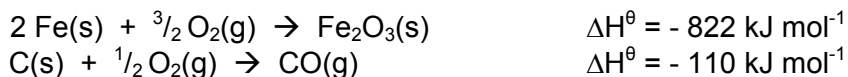


What are the values of angle E and angle F in a molecule of adrenaline?

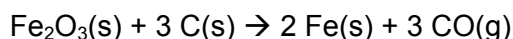
	angle E	angle F
A	120°	107°
B	120°	90°
C	109°	107°
D	109°	90°

Angle E involves the benzene carbon. It is trigonal planar. Hence, 120° bond angle. As for Angle F, it is similar to ammonia. There is a lone pair of electrons on nitrogen so its bond angle is expected to be slightly less than 109°.

- 9** The standard enthalpy changes for two reactions are given by the equations:

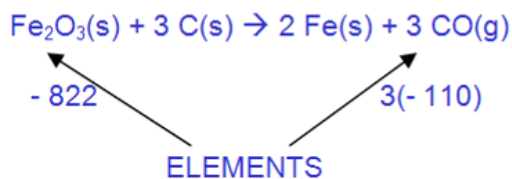


Using the above information, what is the standard enthalpy change for the following reaction?



- A** -712 kJ mol⁻¹
- B** -492 kJ mol⁻¹
- C** +492 kJ mol⁻¹
- D** +712 kJ mol⁻¹

Students need to recognise that the standard enthalpy changes for two reactions given are actually enthalpy changes of FORMATION of the compounds. (1 mole of compound formed from elements in standard state.)

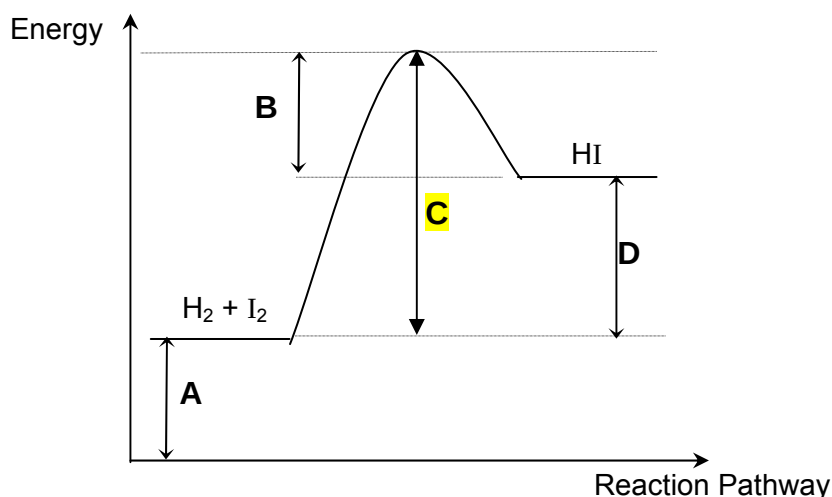


Using Hess' Law, $\Delta H - 822 = 3(-110)$.

$$\Delta H = +492 \text{ kJ mol}^{-1}.$$

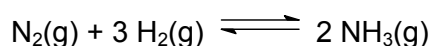
- 10 The reaction pathway diagram for the reaction $\text{H}_2(\text{g}) + \text{I}_2 \rightarrow 2\text{HI}(\text{g})$ is shown below.

Which correctly labels the activation energy of this reaction?



Activation energy is measured from the energy of the reactants to the first peak (in this case, only peak) of the energy profile diagram.

- 11 In the Haber process, ammonia is manufactured according to the following equation:



Which of the following gives the correct condition used in this reaction?

- A 100°C
- B 150 atm
- C Fe catalyst**
- D 1 mol dm⁻³ of H₂

- A Correct temperature range: 450 - 500 °C
- B Correct pressure range: 200 - 300 atm
- D H₂ is a gas, and *concentration* is not used to describe it, rather *pressure* is used instead.

- 12 How much water must be added to 2.0 dm^3 of a solution of pH 2.0 in order to increase its pH to 3.0?

A 1.0 dm^3 B 10 dm^3 **C 18 dm^3** D 20 dm^3

For a solution with pH 2.0,
 $[\text{H}^+] = 10^{-\text{pH}} = 10^{-2} \text{ mol dm}^{-3}$
 amount of H^+ ions in $2.0 \text{ dm}^3 = 10^{-2} \times 2 = 0.02 \text{ mol}$

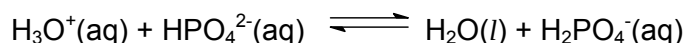
For a diluted solution with pH 3.0, the amount of H^+ remains unchanged,
 let the total volume be $x \text{ dm}^3$,

$$0.02/x = 10^{-3}$$

$$x = 20.0 \text{ dm}^3$$

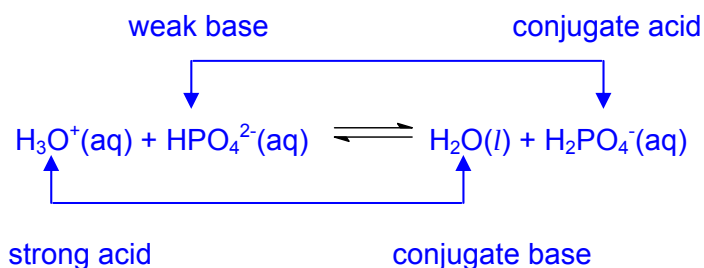
Hence, volume of water added = $20.0 - 2.0 = 18.0 \text{ dm}^3$

- 13 For the equilibrium shown below,



which of the following is a conjugate Bronsted-Lowry acid-base pair?

	Base	Conjugate Acid
A	HPO_4^{2-}	H_3O^+
B	HPO_4^{2-}	H_2PO_4^-
C	H_3O^+	H_2O
D	H_3O^+	H_2PO_4^-



- 14 The rate of reaction between a strip of magnesium metal and 50.0 cm^3 of 1 mol dm^{-3} dilute hydrochloric acid is determined at 25°C .

In which of the following cases would both sets of conditions bring about an increase in the reaction rate?

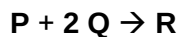
- A Magnesium powder and 100 cm^3 of 1 mol dm^{-3} dilute hydrochloric acid.
 B Magnesium powder and 50.0 cm^3 of 0.8 mol dm^{-3} dilute hydrochloric acid.
 C 100 cm^3 of 1.0 mol dm^{-3} dilute hydrochloric acid at 30°C .
D 50.0 cm^3 of 1.2 mol dm^{-3} dilute hydrochloric acid at 30°C .

Reaction rate depends on *temperature*, *surface area* and *concentration of reactants*.

Reactants: Mg metal, 0.05 mol of HCl

- A Increase in *surface area*, but *concentration* of HCl remains unchanged.
 B Increase in *surface area*, but *concentration* of HCl is lower.
 C Increase in *temperature*, but *concentration* of HCl remains unchanged.
 D Increase in both *temperature* and *concentration* of HCl .

- 15 Two substances **P** and **Q** react according to the following equation:



To determine the order of reaction with respect to each substance, four experiments were performed and the results are shown below:

Experiment	[P]/mol dm ⁻³	[Q]/mol dm ⁻³	Initial rate of formation of R/ mol dm ⁻³ min ⁻¹
1	0.10	0.10	0.0010
2	0.40	0.10	0.0160
3	0.10	0.20	0.0020
4	0.20	0.20	x

Which of the following gives the correct value of x?

- A 0.0040 **B 0.0080** C 0.0160 D 0.0320

By observations, using Expts 1 and 2,
when [B] is constant, when [A] increases by 4x, rate increases by 16x, hence order of reaction w.r.t. A is 2.

By observations, using Expts 1 and 3,
when [A] is constant, when [B] increases by 2x, rate increases by 2x, hence order of reaction w.r.t. B is 1.

$$\text{rate} = k [\text{A}]^2 [\text{B}]$$

Using Expts 1 and 4,

$$\frac{0.0010}{\text{rate}_1} = \frac{k(0.10)^2(0.10)}{k(0.20)^2(0.20)} = 0.0080 \text{ mol dm}^{-3} \text{ min}^{-1}$$

- 16 Elements **X**, **Y** and **Z** react with oxygen to form their respective oxides.

When dissolved in water, oxide of **X** gives a strongly alkaline solution while oxide of **Z** gives a strongly acidic solution. Oxide of **Y** is able to react with both acids and bases.

Given that elements **X**, **Y** and **Z** are Period 3 elements, which of the following is **not** true for these three elements?

- A The atomic radius decreases from **X** to **Y** to **Z**.
 B The electronegativity increases from **X** to **Y** to **Z**.
 C The first ionisation energy increases from **X** to **Y** to **Z**.
D X, Y and Z could be sodium, magnesium and aluminium respectively.

P is an alkaline oxide → is Na₂O (note MgO is sparingly soluble and only weakly basic)

R is an acidic oxide → is either P₄O₆/P₄O₁₀ or SO₂/SO₃ (note SiO₂ is insoluble in water despite it being an acidic oxide)

Q is an amphoteric oxide → is Al₂O₃

Hence, **X**, **Y** and **Z** are Na, Al and P/S.

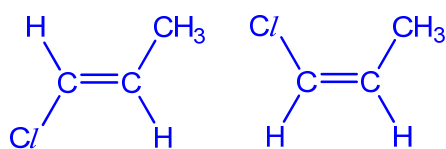
17 Which of the following is **not** true about phosphorus?

- A It has a lower melting point than sulfur.
- B It can exhibit +3 and +5 oxidation states.
- C It can form chlorides that hydrolyse in water.
- D It can form oxides with giant covalent structures.**

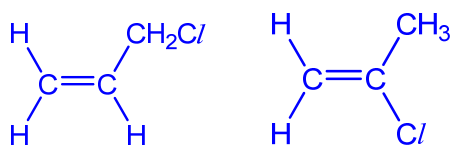
- A P exists as P_4 while S exists as S_8 (higher M_r), hence VDW forces is weaker in P, and it has a lower melting point.
- B True eg PCl_3 and PCl_5 .
- C True
 $PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$
 $PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$
- D False, both P_4O_6 and P_4O_{10} have simple molecular structures.

18 What is the total number of isomers (including stereoisomers) for a non-cyclic organic compound, C_3H_5Cl ?

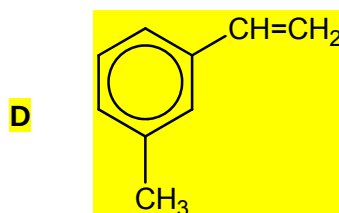
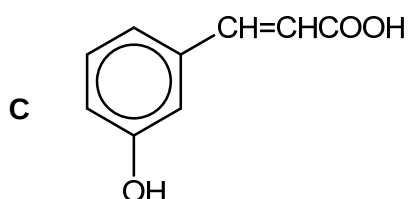
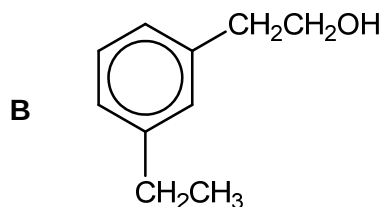
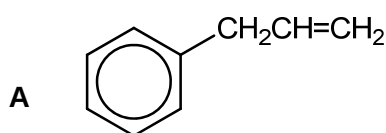
- A 2 B 3 **C 4** D 5



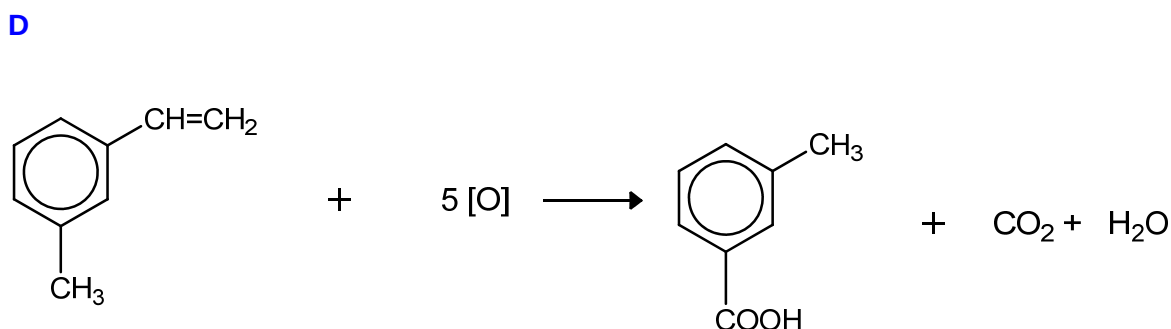
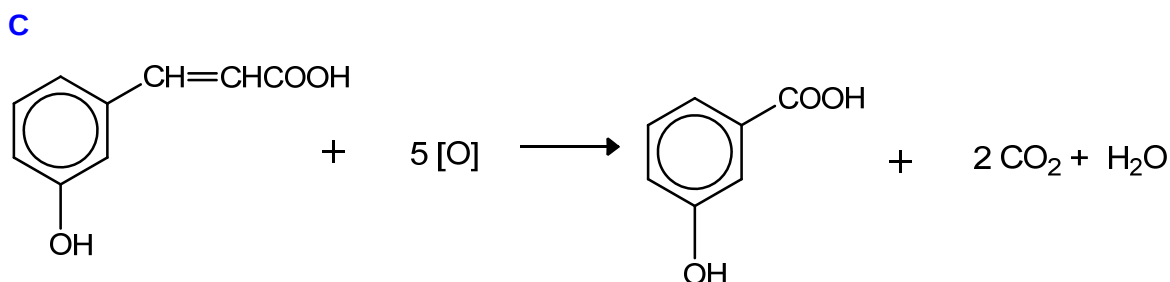
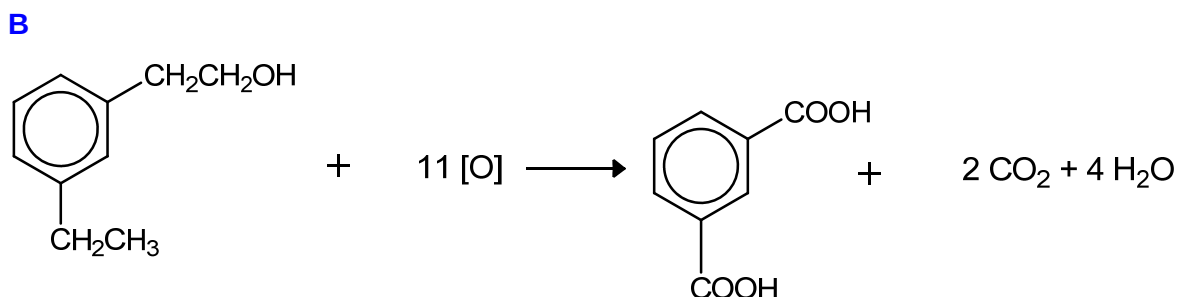
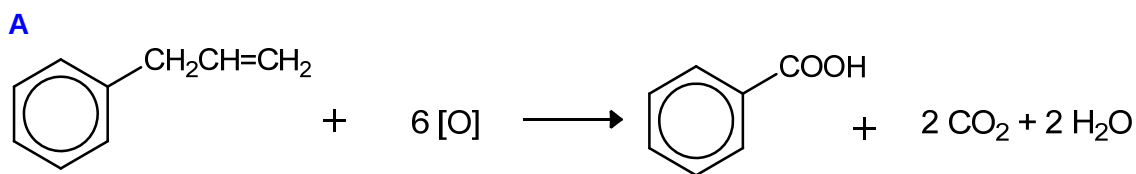
(These 2 are geometric isomers)



19 Which one of the following will **not** liberate 2 moles of carbon dioxide when 1 mole of the compound is treated with excess hot acidified potassium manganate(VII)?



Regardless of the chain length of the alkyl group, the alkylbenzene will be oxidised to benzoic acid.



- 20 Which of the following reagents may be used to distinguish between 1-bromobutane and 2-bromobutane?

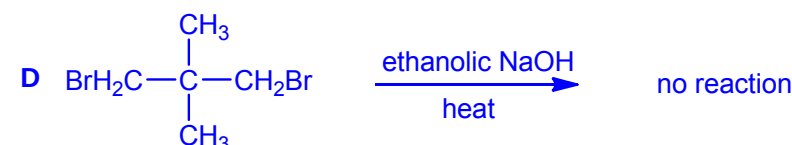
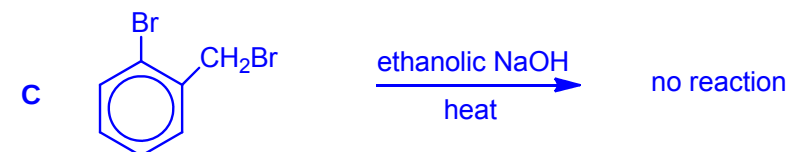
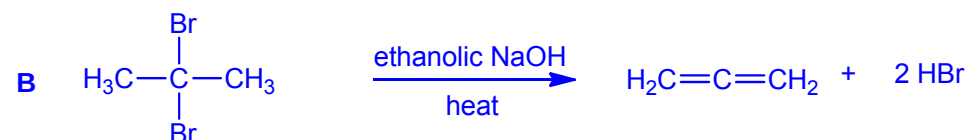
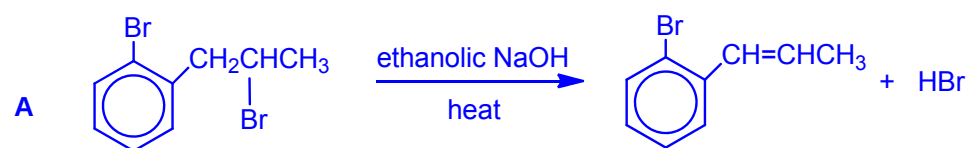
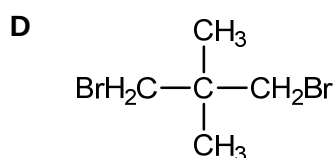
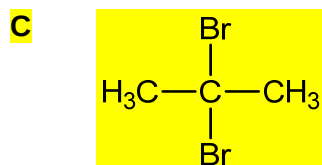
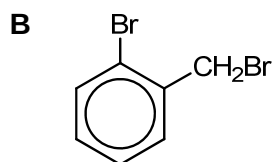
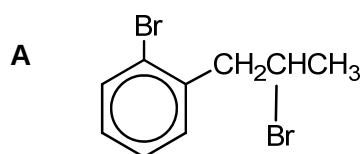
- A** alkaline aqueous iodine
B sodium hydroxide
C acidified potassium manganate(VII)
D aqueous silver nitrate

In the presence of NaOH(aq) and heat, 1-bromobutane will become butan-1-ol, but this will not react further with the iodine.

On the other hand, after the substitution of 2-bromobutane, butan-2-ol will further react with the aqueous iodine to give a yellow precipitate of CHI₃.

- 21 1 mole of an organic compound **P** undergoes a reaction with alcoholic sodium hydroxide to form 2 moles of HBr.

Which of the following could be **P**?



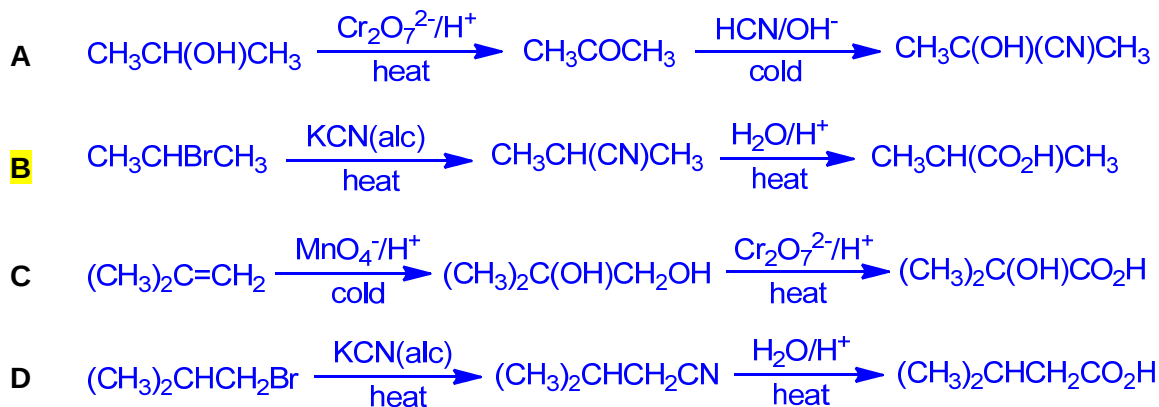
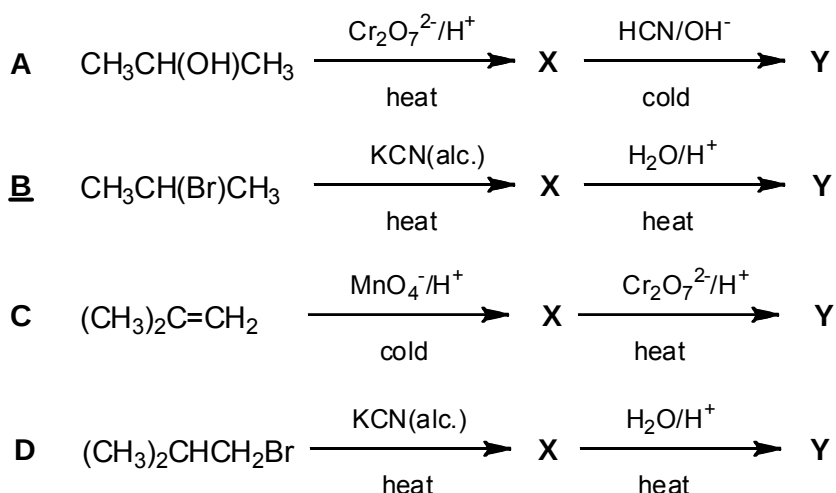
- 22 A student was tasked to determine the halogen (either Cl, Br or I) present in a sample of liquid halogenoalkane. He added 1 cm³ of the sample to 1 cm³ of aqueous NaOH in a test-tube, followed by heating the test-tube in a water bath for 5 minutes. He then added 1 cm³ of dilute HCl to neutralise any excess NaOH. Finally, he added 3 drops of aqueous AgNO₃ and observed the colour of the precipitate formed.

What was the error in the student's experiment?

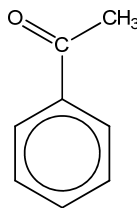
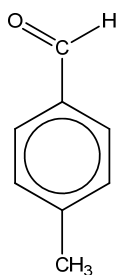
- A** using 1 cm³ of aqueous NaOH
B heating for 5 minutes
C using 1 cm³ of dilute HCl
D using 3 drops of aqueous AgNO₃

Dilute HNO₃ should be used to neutralise any excess NaOH. HCl will introduce Cl⁻ into the solution and a white ppt of AgCl will definitely form upon addition of AgNO₃. This gives the test a false positive for the presence of chloride.

23 Which of the following syntheses will produce compound Y, 2-methylpropanoic acid?



24 Which of the following chemical tests will give different observations for the following two compounds?



- A Warm with 2,4-dinitrophenylhydrazine
 B Warm with Fehling's solution
 C Warm with Tollen's reagent
 D Warm with aqueous bromine

- A Aldehyde group and ketone group will both react.
 B Aromatic aldehyde and ketone will both NOT react with Fehling's.
 C Correct answer.
 D Both do not have alkene group, so both will not react.

25 Consider the following four compounds:

1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
2. $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
3. $\begin{array}{c} \text{CH}_3\text{CHCO}_2\text{H} \\ | \\ \text{F} \end{array}$
4. $\begin{array}{c} \text{CH}_3\text{CHCO}_2\text{H} \\ | \\ \text{Cl} \end{array}$

What is the order of increasing acidity of the compounds (weakest to strongest)?

- A** 1 → 2 → 4 → 3
B 1 → 2 → 3 → 4
C 3 → 4 → 2 → 1
D 3 → 4 → 1 → 2

Compound 1 is an alcohol while the other three are acids. So, compound 1 will definitely be the weakest in terms of its acidity.

Among the other three compounds, compound 2 is propanoic acid. Compounds 3 and 4 have electron withdrawing substituents on propanoic acid, and will thus be more acidic than propanoic acid itself. Since F is the stronger electron withdrawing group compared to Cl, Compound 3 it will polarize the O-H bond most strongly and hence cause this acid to be the most acidic.

For questions 26 to 30, one or more of the three numbered statements 1 to 3 may be correct. The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

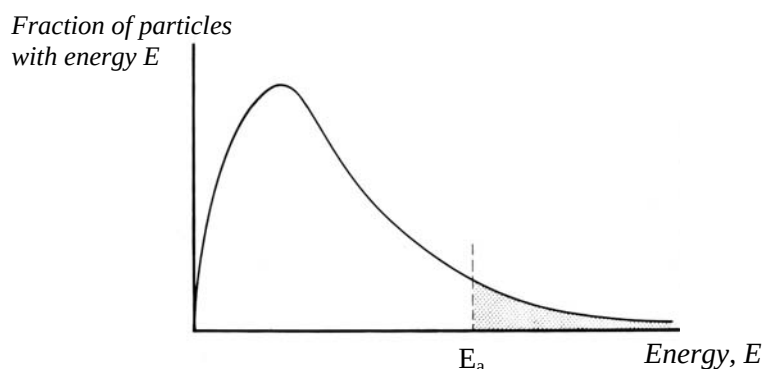
No other combination of statements is used as a correct response.

26 Which of the following substances conduct electricity due to delocalised electrons?

- 1** graphite
2 benzene
3 molten sodium chloride

- 1** Benzene has delocalized electrons only within the ring and they do not make benzene a conductor of electricity.
3 Molten sodium chloride can conduct electricity but through mobile ions, and not delocalized electrons.

- 27 The diagram below shows the Boltzmann Distribution of molecular energies at a particular temperature.



When the temperature is increased, which of the following occurs?

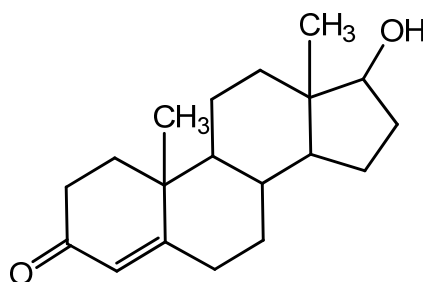
- 1 The activation energy is lowered.
 - 2 The graph becomes broader.
 - 3 The peak shifts to the right.
- 1 The activation energy remains unchanged.
NOTE: activation energy is lowered **only** if a catalyst is added.

- 28 Germanium is directly below silicon in the Periodic Table.

Which statements about germanium are **true**?

- 1 It exists as a solid at room temperature.
 - 2 It has a smaller ionic radius than silicon.
 - 3 It does not conduct electricity at all temperatures.
- 2 It has a larger ionic radius than silicon because of increased shielding effect.
- 3 It only conducts electricity under certain temperatures.

- 29 Testosterone is one of the most important male sex hormones and its structure is given below.



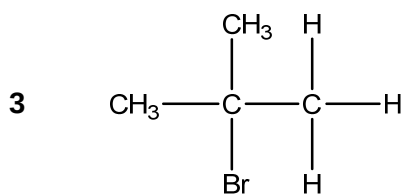
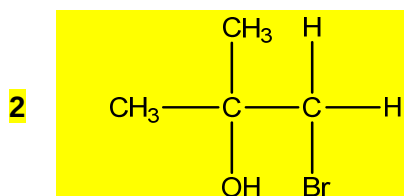
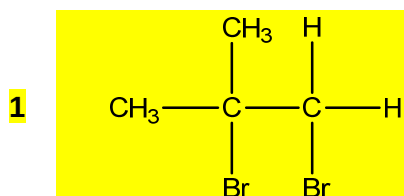
Testosterone

Which of the following statements are true?

- 1 It gives an orange precipitate with 2,4-dinitrophenylhydrazine.
- 2 It decolourises orange aqueous Br_2 .
- 3 It reacts with SOCl_2 to give steamy fumes of HCl .

- 1 Testosterone contains a ketone group.
- 2 Testosterone contains a C=C double bond, and addition occurs readily.
- 3 Testosterone contains a secondary alcohol, and substitution occurs.

30 Which of the following are possible products when 2-methylpropene is reacted with aqueous bromine?



Option 3 will only be formed when 2-methylpropene is reacted with HBr and not Br₂.